

Rank 1 Constraint System

Def: R1CS is an arithmetic circuit with requirement that each equality constraint has one multiplication.
(No constraint on number of additions)

for Eg:

$$x = nyzu$$

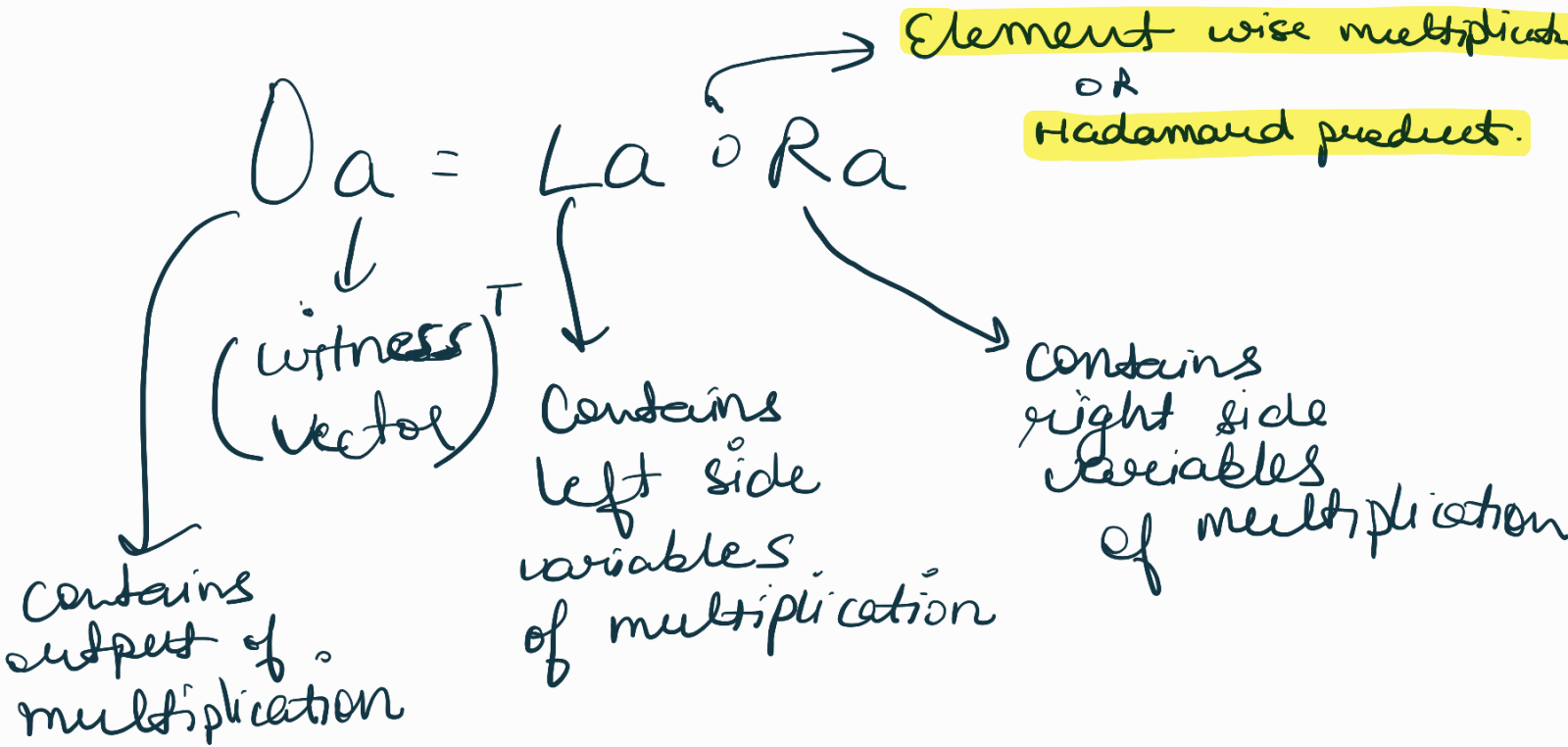
$$v_1 = ny$$

$$v_2 = zu$$

$$x = v_1 v_2$$

witness vector =

$$\begin{bmatrix} 1 \\ x \\ y \\ z \\ v_1 \\ v_2 \\ x \end{bmatrix}^T$$



• Witness vector is $1 \times m$

↓

no. of inputs +
outputs + intermediates
+ 1

• L , R and D are matrices with same number of columns as witness vector.

• L , R and D have size $n \times m$.

Where n : no. of constraints

m : no. of variables + 1

✓
as we add one
constant 1.

So in our example, $m = 3$

$$n = 8$$

$$\text{Witness vector} = [1 \ x \ n \ y \ z \ u \ v_1 \ v_2]$$

Constraints : $v_1 = x \times y$

$$v_2 = z \times u$$

$$\underbrace{x}_{\substack{\downarrow \\ \text{Output} \\ \text{term}}} = \underbrace{v_1}_{\substack{\downarrow \\ \text{LHS}}} \times \underbrace{v_2}_{\substack{\downarrow \\ \text{RHS}}}$$

$$L = \begin{bmatrix} 1 & x & n & y & z & u & v_1 & v_2 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$R = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$O = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Guess : $z = n * y + 2 \Rightarrow z - 2 = n * y$

$$n = 1$$

$$m = 3$$

$$\text{Witness vector} = [1 \quad n \quad y \quad z]$$

$$L = [0 \quad 1 \quad 0 \quad 0]$$

$$D = [-2 \quad 0 \quad 0 \quad 1]$$

$$R = [0 \quad 0 \quad 1 \quad 0]$$

In short we are writing constraints as :

for eg : $z = n * y$

$$(0 * 1 + 0 * n + 0 * y + 1 * z) = (0 * 1 + 1 * n + 0 * y + 0 * z) * (0 * 1 + 0 * n + 1 * y + 0 * z)$$

NOTE : Everything is done modulo prime in RLCS.

• What if there is no multiplication.

Ex : $z = n + y$

Rewrite it as $z = (n + y) * 1$